

## THE BUILD

# Whose HLA is in your reference?

Five projects. One graph of the MHC, a highly variable immune region.

# 754

## MHC haplotypes

390 Asian or Arab haplotypes

752 biological haplotypes + 2 references

APR / HPRC r2 / JaSaPaGe / K-PanRef / CPC



Extract MHC → annotate → build graph (schematic)

### A SHARED RESOURCE

Graph + allele labels

### A WORKING PIPELINE

CWL workflow

### A TEST BED

Reads: 2,504 samples

## THE PAYOFF

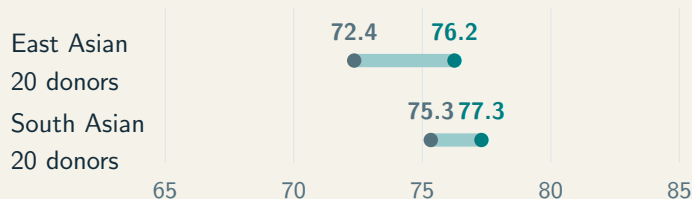
# More haplotypes. Better SV calls.

Better recovery of genotypes carrying structural variants.

## SV-bearing genotypes recovered exactly (%)

● HPRC-only panel

● Expanded panel



**+3.9**

points in East Asian donors

95% CI: +2.4 to +5.3

South Asian: +2.0 points

95% CI: +0.3 to +3.5

## The clearest gain: recovering harder variants.

40-donor pilot, PanGenie, same reads; assembly-derived truth. Panel size and ancestry change together.

Test families excluded from panels; test assemblies remain in graph topology. All eligible SV-bearing genotypes.

## THE SURPRISE

# Better sequences. Same names.

A small change to SpecHLA: add pangenome sequences when collecting reads.

### Exact gene-body matches (out of 128)

8 donors  $\times$  8 genes  $\times$  2 haplotypes; assembly truth

Standard SpecHLA

41

+ pangenome in read binning

51

10 more exact sequences; same allele calls (61/63).

LOCITYPER / 40 DONORS

### T1K leads on names.

Classical genes, 40 donors:

T1K **98.4%** / our panel **92.5%**

DRB3/4/5 type + copy number:

our panel **111/113 correct**  
Stage B; before exon grafting.

## Next: take this beyond the pilot.

Locked validation: 228 leave-one-out donors + 946 experimentally typed donors.